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I confirm that I understand my coursework needs to be submitted online via MST Classroom under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

Acknowledgement

I would like to greatly appreciate my outstanding supervisors Mr. Prashant Pudasaini and Mr. Suraj Neupane who have supported me in every way possible at any stage of this project and helped me out.

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Abstract

The Intelligence Recon System (IRS) is an AI-based cyber security reconnaissance platform that will automate and centralize the vulnerability assessment processes. It combines several command-line security tools into one, web based interface, which allows asynchronous, parallel execution to speed up and more efficient scan. An AI decision node is a smart tool that recommends and orders tools depending on target context and user intent, and minimizes the influence of a human bias in the triage process and configuration errors. The system structures raw outputs and aggregates them to produce easy to read and understand data, which facilitates the analysis of results and simplifies the work of security teams. A live dashboard is used to show the progress of the scans, assets found and vulnerabilities discovered and persistent storage stores the entire historical data to be analyzed and audited. Standardized, shareable reports are also created by the IRS to assist in complying, recording of incidents, and communicating with stakeholders. Replacing the manual recon with an automated intelligent system, the platform enhances the organizational security posture and enables advanced reconnaissance to be available even to smaller teams.

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Abbreviations	Full Form
Intelligence Recon System	IRS
Command Line Interface	CLI

1. INTRODUCTION

Cyber security reconnaissance is an essential step in the cyber security sector, where systematically collecting information about target systems, networks and organizations takes place (Cymulate, 2025). This is done to detect any vulnerabilities, network structures and network security measures which can be used to attack in the case of a cyber-attack. Reconnaissance helps threat actors to develop detailed profiles of targets to strategize effective attack methods, and defenders to use such techniques on the offensive to determine and strengthen security positions (Arora, 2025).

1.1. Introduction to Topic

When it comes to the security assessment, the use of various disparate tools which must be manually executed to provide reconnaissance data is time-intensive and difficult to implement. By automating these processes using the **IRS**, a single framework, the efficiency of the process improves, the complexity of the user process is minimized, and the presentation of the data becomes uniform. Automation combined with organized output makes it easier to identify vulnerabilities and make informed decisions, which is essential in the modern dynamic environment of a threat.

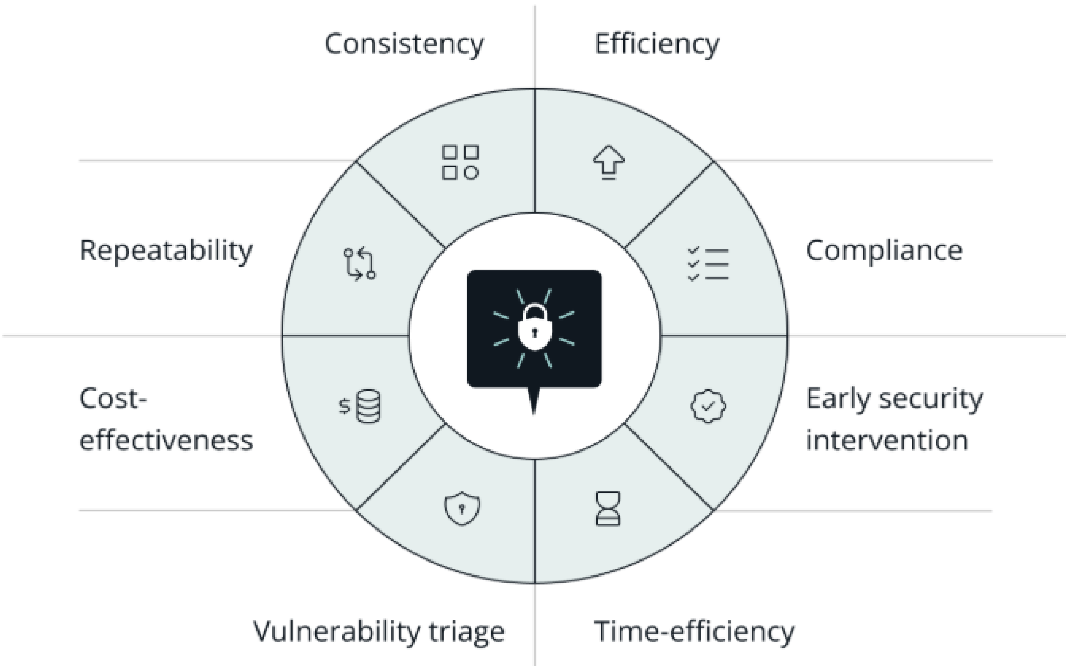

Figure 1 Benefits of Automated Security Testing (Sasovets, 2025).

1.2. Problem Statement

1. **Manual and Time Consuming:** Manual recon is time-consuming, failing to keep up with changing cyber threats and demand of timely security opinions. (Sumanth, 2023).

2. **Human Error and Bias:** Even professional ethical hackers have a tendency to make errors, such as failing to interpret information or not detecting a vulnerability. False alarms or missing threats can be caused by unconscious bias.
3. **Manual and Fragmented Execution:** Recon involves chaining numerous manual CLI tools. This fragmentation wastes time in complicated syntax and leads to inefficiency and inconsistency in the outcomes (Ariganello, 2022).
4. **Absence of Centralized Management and Professional Reporting**: Scan results are distributed locally thus making it difficult to retrieve the data and track its history. Report generation is slow and manual, and prone to errors.

1.3. Project as Solution

1. **Removes Manual Time Waste:** The system automates the whole working system allowing the security analysis to take place in a fast and timely manner, even in extensive settings.
2. **Reduces Minimizes Error and Bias:** The inclusion of AI Decision Node makes the tool selection objective and optimized, which greatly reduces human misinterpretation or bias when making triage decisions.
3. **Automated and Parallel execution:** This system incorporates the specialty CLI tools into one scalable framework, permitting automated and parallel chaining to conquer the disjointed and manual execution.
4. **Centralized Management and Reporting:** Data is gathered, normalized, and stored continuously, to give consistent and structured outputs to real-time analysis and the creation of professional reports to the stakeholders with a single click.

2. Aim and Objectives

2.1. Aim

The Aim of the IRS is to create, design, and thoroughly test an integrated, unified, and smartly coordinated security platform that automates the multi-step, multi-stage reconnaissance process of professional CLI programs, and thus replaces manual and fragmented vulnerability testing with automated, centralized, and report-enabled security testing based on the OWASP Top 10 threats.

2.2. Objectives

To achieve the project goal, the following objectives that are need to be achieved:

- Created a strong, 3-tier (Frontend, Fastapi Backend, and Tool Execution) web and Python-based platform.
- Develop and implement one easy-to-use web dashboard to minimize inputs, provide real time response, and enable OWASP-oriented target and tool selection.
- Have asynchronous execution with dynamic chaining of different CLI tools with a reliable data flow and a responsive non-blocking user interface.
- Add an AI Decision Node that is used to optimize the tool execution plan contextually by smartly utilizing the possibility of excluding irrelevant or redundant scanning checks.
- Create an effective data parsing component to process all raw and inconsistent tool output and provide this in a centralized, consistent and structured format.
- Apply persistent storage to provide all historical scan results enabling users to monitor the history of asset discovery and vulnerability.
- Offer a real time web dashboard with scans and the capacity to create and download professional, standardized PDF/HTML reports.

3. Expected Outcomes and Deliverables

The purpose of the IRS project is to replace manually executed, time-intensive command lines to a singular, user-friendly system driven by AI-based intelligence. Enhancing vulnerability assessment process is done through automating the process of selecting and executing security tools according to the real-time context of the situation and the intent of the user, this reduces the amount of human effort and maximizes efficiency.

3.1. Deliverables

1. **IRS**

   The complete system can execute hyper-focused, AI-optimized security assessments via one web screen.

2. **Unified Interface and Monitoring Dashboard**

   This allows analysts to manage all security scans, track real-time progress, and review aggregated findings from a single central hub.

3. **Structured Output and Persistent Results**

   It transforms raw data so that the user can easily analyze the standardized results and securely store all the historical data for long-term review.

4. **Security Tool Execution Layer**

   This layer runs the 15 in-built tools, enabling the system to execute a complete and powerful suite of security checks instructed by the AI.

5. **Downloadable Standardized Report**

   Users can generate professional, ready-to-share HTML reports for audits, documentation, or incident review.

4. Project risks, threats and contingency plans

Threats and risks can be encountered during the project development process, since every single system can be exposed to risks. The following risks and threats are mitigated along with contingency plans in order to minimize systemic defects.

- **AI Triage Dependency Risk**

  Risk: AI Triage (GeminiAPI) provides faulty filtering, which reduces the effectiveness of scans.

  Threats: GeminiAPI update software failure, AI misunderstands target, or proposes a tool that is not relevant.

  Contingency plan

  - Introduce Manual Override and Fallback to operate 5 most important tools without AI.
  - Have a strict version locked prompt template.

- **Creation of tool Risk and Data Inconsistency risk.**

  Risk: Security tools malfunction, crash system or invalid output of raw results.

  Threats: Critical tool will not run with update, change of format output or tool requires unconfigured environmental variables.

  Contingency plan

  - Install Version Locking and Isolation on all the 15 security tools.
  - Crash of subprocesses are isolated using the try/except in Python.
  - In case of parsing failures, do away with inconsistent data and record the error.

- **Performance and Scalability risk.**

  Risk: Asynchronous Execution Core is inefficient in managing concurrent tools, thus leading to system lag.

  Threats: Subprocess module will be a bottleneck, server will run out of resources or database will write slowly.

  Contingency plan

  - A Fixed Parallel Limit will be implemented to control simultaneous tools.
  - Make use of the inbuilt asyncio of Python to have a responsive UI.

5. Methodology

Methodology refers to the comprehensive strategy devised for executing a project. SDLC is a guided process which is applied to design, develop and test high quality software (GeeksForGeeks, 2025). *For more details*

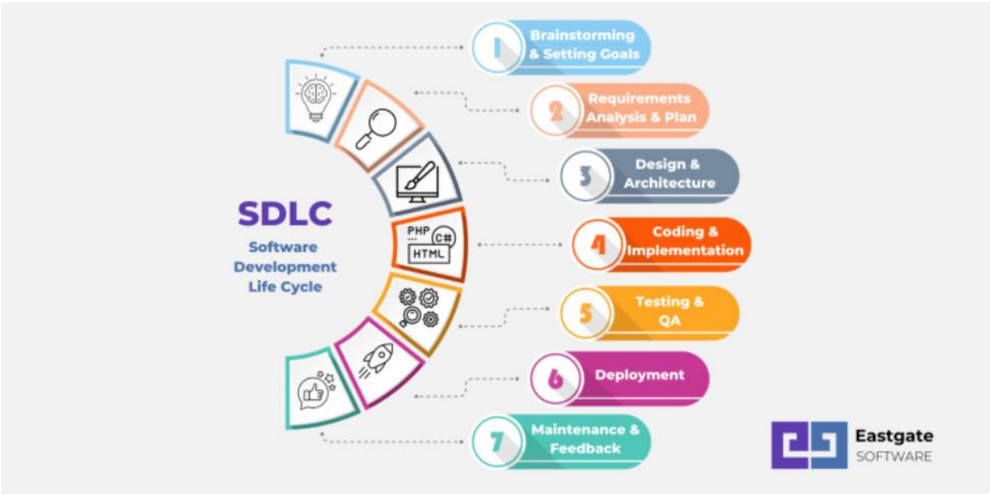

Figure 2 Software Development Lifecycle (Eastgate, 2025).

5.1. Considered Methodology

In IRS, the risk mitigation required the proper choice of SDLC which has led to the option of both Waterfall and Agile approaches.

5.1.1. Waterfall methodology

It's a linear process consisting of five steps namely requirements, design, implementation, verification and maintenance (Day, 2025). *For more details*

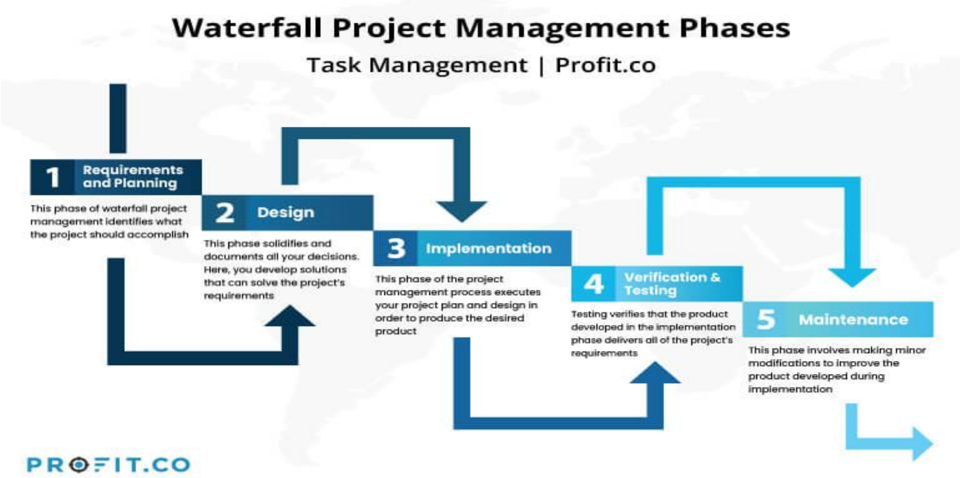

Figure 3 Waterfall Methodology (Welch, 2025).

5.1.2. Agile Methodology

It refers to a project management model that divides projects into a number of ongoing processes, otherwise referred to as sprints (Peek, 2024). *For More Details*

Figure 4 Agile Methodology (Kamis, 2019).

5.1.3. V-Model Methodology

V-Model is a methodology which combines both Verification (design and development) and validation phases, creating a V-shape in which testing phases resemble development ones (GeeksForGeeks, 2025). *For more details*

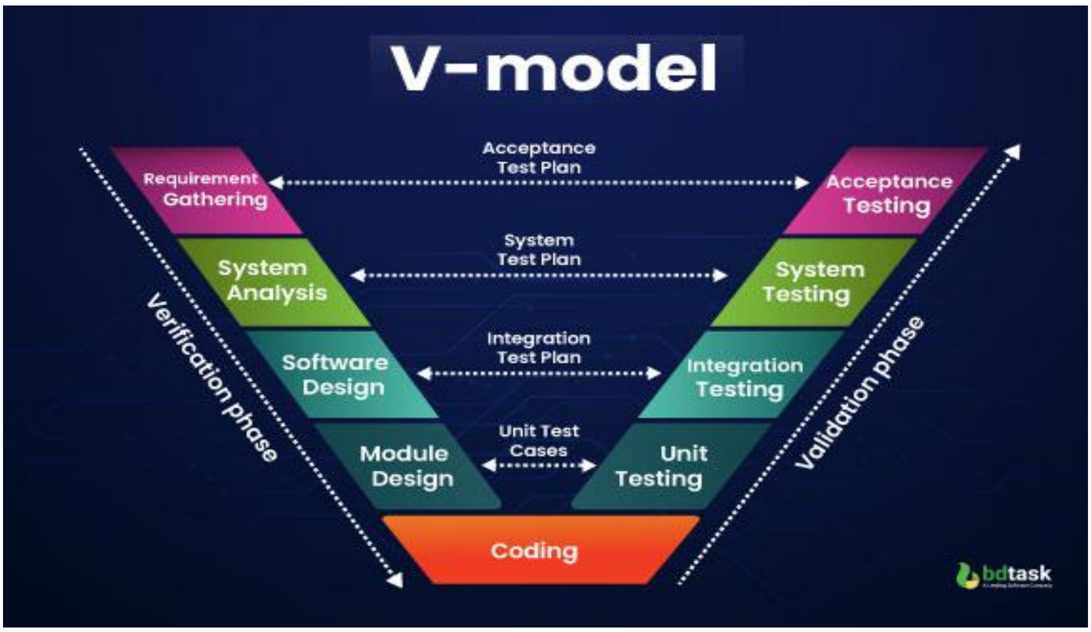

Figure 5 V-Model Methodology (Simran, 2025).

5.2. Selected Methodology

After defining essential project attributes, the selected management system is the Agile Scrum Methodology that focuses on iterative development, flexibility, and collaboration.

5.2.1. Agile Methodology-Scrum Methodology

Agile Scrum Methodology is officially chosen to be the project management tool used in this IRS project. This option has been informed by the fact that it requires a very flexible strategy in dealing with challenging regulatory requirements and a fast changing specification.

The framework is based on the three pillars to make it controlled and transparent:

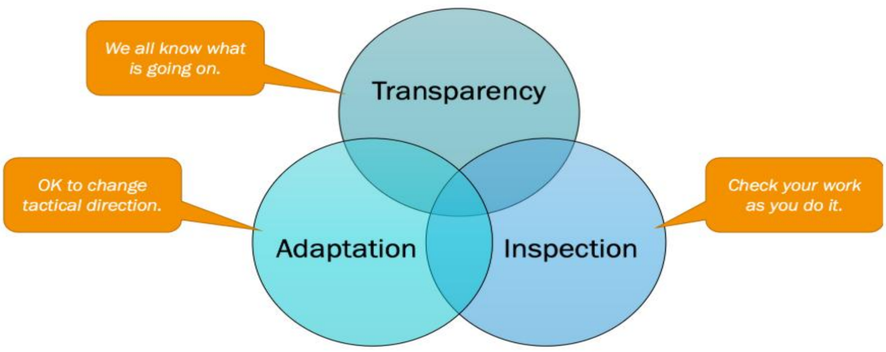

Figure 6 Three Pillars of Scrum (Doshi, 2016).

- **Transparency**

  Being transparent in terms of all facts, progress, and purposes to be shared among all individuals engaged to create a shared understanding and establish trust.

- **Inspection**

  Consistently checking the product and processes by both the team and stakeholders to identify the unwanted variances or problems.

- **Adaption**

  It is a continuous process of improvement and changes in response to the outcome of the inspection in order to maximize value and performance.

This are the 3 pillars of Scrum (Doshi, 2016).

6. Resource Requirements

6.1. Hardware Requirement

Personal device will be used as development machine.

6.2. Software Requirement

Category	**Software Requirements**	**Specifications**
OS	Ubuntu	Open-source and stable Linux distribution as the development environment.
Development Tools	VS Code	Mainly used IDE to code, debug and version control.
	HTML, CSS, JavaScript	Technologies of User Interface developments.
	Python/FastAPI	Python backend language: Fastapi, high performance, asynchronous API.
Core AI Model	Gemini2.5 Flash	Intelligent triage and data validation of the AI Decision Node
Data Storage	PostgreSQL or Firestore	Storage of all scan results and history record keeping.
Gantt Chart	Clickup	Project timeline and progress tracking visual tool.
Documentation	MsWord	Keep records of project.
Diagram	Eraser, Draw.io	Shows ideas and process.
Version control	Git	Tracks the changes to the source file
Project Management Tool	Trello	Used to keep track of milestones

Table 1 Table of Software Requirements

7. Work breakdown structure

The Work Breakdown Structure illustrated below separates the project into distinct phases and tasks, facilitating the organization, scheduling, and management of all work required for successful completion

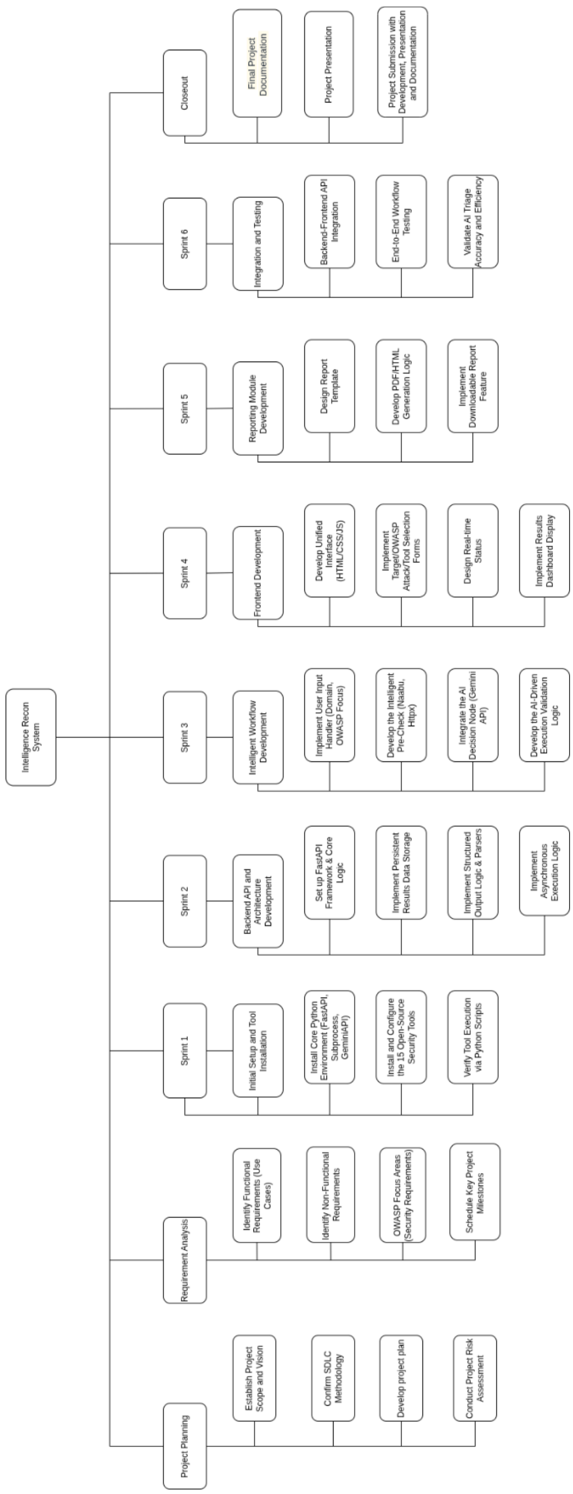

Figure 7 Work Breakdown Structure

8. Milestones

Expected Date	Milestone	Description
November-21, 2025	Project Approval	Project Topic was finalized to be “Intelligence Recon System”.
December-3, 2025	Proposal Submission	Submit proposal outlining the project’s goals, expected results, and risk plan.
December-15, 2025	Backend Setup	Setup the Python environment and tools ready.
December-30, 2025	Backend Core Development	Backend built, main logic working.
December-30, 2025	Interim Report Submission	Interim report with project progress.
January-31, 2026	Frontend and Dashboard	Create user interface and dashboard.
April-8, 2026	Reporting Integration and Testing	Add reports, connect parts, and test system.
April-22, 2026	Final Report and Artifacts Submission	Hand in report and all deliverables.

Table 2 Project Milestone

9. Project Gantt chart

1. Project Planning

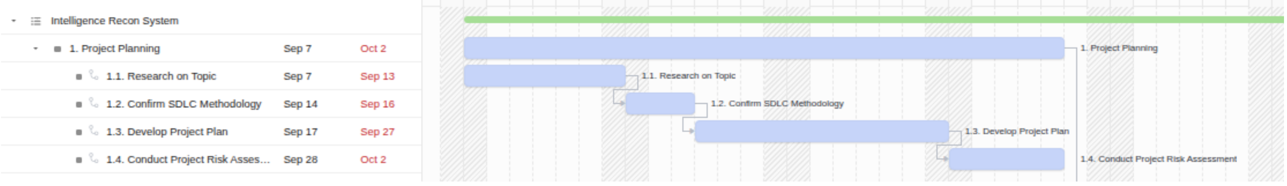

Figure 8 Project Planning

2. Requirement Analysis

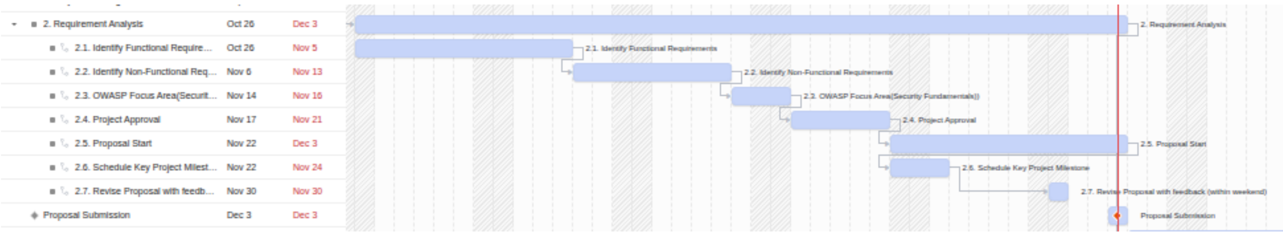

Figure 9 Requirement Analysis

3. Sprint 1

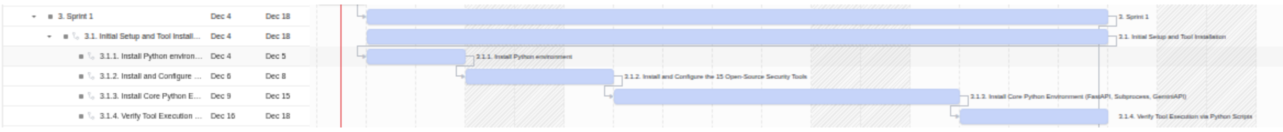

Figure 10 Sprint 1

4. Sprint 2

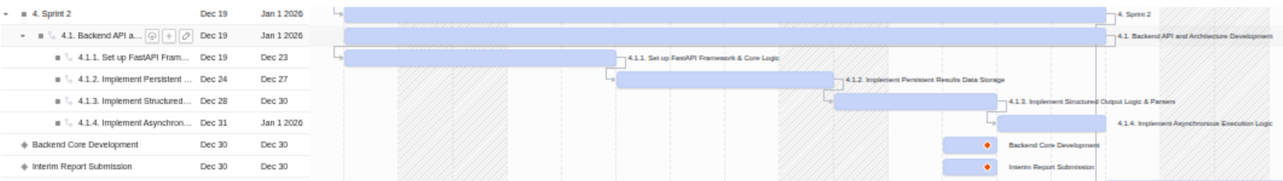

Figure 11 Sprint 2

5. Sprint 3

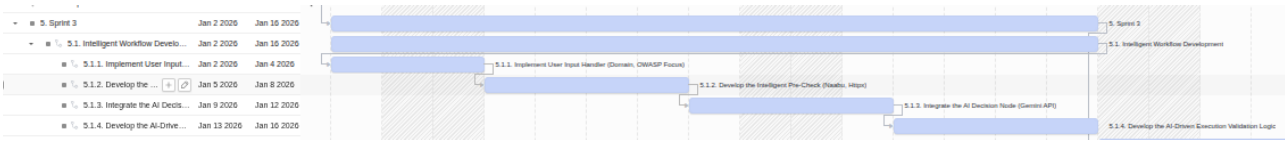

Figure 12 Sprint 3

6. Sprint 4

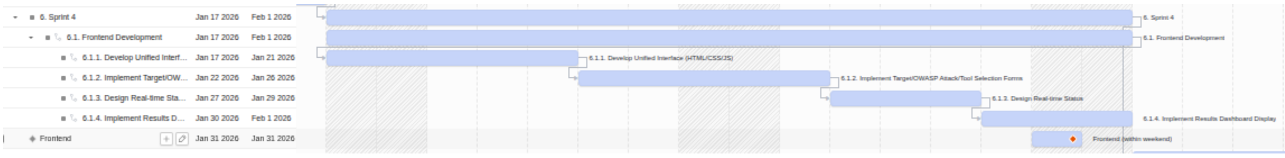

Figure 13 Sprint 4

7. Sprint 5

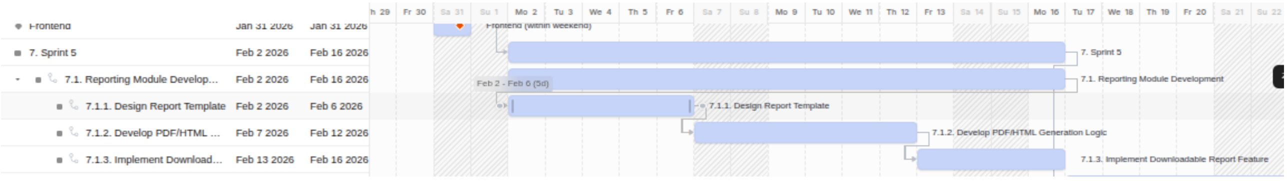

Figure 14 Sprint 5

8. Sprint 6

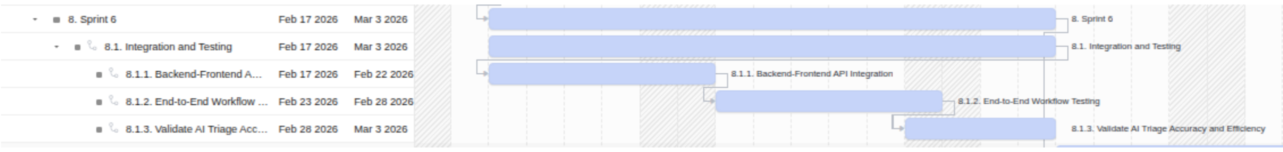

Figure 15 Sprint 6

9. Control

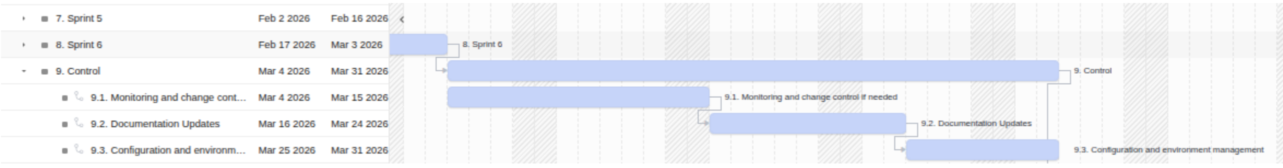

Figure 16 Control

10. Closure

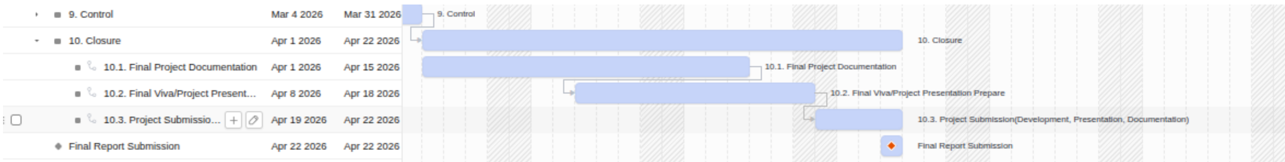

Figure 17 Closure

10. Conclusion

The Intelligence Recon System project provides an automated, AI-based reconnaissance system that will substitute disjointed command-line processes with a single, web-based system. Its integration of several security tools, asynchronous jobs, and an AI decision node simplify the vulnerability evaluation process, human error reduction, and manual labor. By having centralized data storage, structured reporting and an agile scrum-based development methodology, the system increases the efficiency, scalability, and usability by security analysts in contemporary cyber defense settings.

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11. Appendix

11.1. Methodology

Methodology is the total structure that shows how a project is going to be done from the initial stage to the final one. It explains the main route (Agile Scrum, Waterfall, and V-Model), the order of the phases, and the detailed activities, instruments, and methods utilized at each level. Furthermore, it allocates the description of the roles and responsibilities, the decision-making criteria, the documentation standards, and the way progress, quality, and risks will be monitored. Since it is made explicit and agreed upon beforehand, methodology changes the indefinite concept ("we will build this system") into a distinct, reusable process that any skilled team could follow and understand (Cobrief, 2025).

Why this is important for a project:

1. It provides an obvious roadmap: everyone should know the first stage (such as requirements), the next one (design, implementation, testing, deployment), and what can be obtained at each step.
2. It makes the work organized and less chaotic: rather than each individual having their own way of doing things, the entire group takes the same steps and gets the same set of standards and this removes confusion and duplication.
3. It enhances control of time cost and quality: since processes, tasks and milestones are established, estimation of effort, delays, scope variation and quality maintenance become easier.
4. It facilitates communication and accountability: since there is a common methodology, one can easily impress supervisors and other stakeholders about what has been done, what is being done and what will be done.
5. It helps make results more credible and professional: once a project has been conducted using a well-established methodology, it can be expected that the work was conducted in a rigorous and verifiable manner and not through trial and error.

This are the important points that shows why methodology are important for the project (Cobrief, 2025).

Click here to go back.

11.1.1. Waterfall Methodology

Waterfall methodology refers to the linear project approach that is executed in phases where each phase (requirements, design, implementation, testing, deployment, and maintenance) is accomplished to completion which is followed by the subsequent one. It is based on high level upfront planning and documentation and is less flexible in terms of scope modification after development has begun. This is practical in stable and predictable projects but not practical in dynamic or research driven work (Day, 2025).

Why Waterfall was considered.

- Simple and easy to follow, having a definite line of stages.
- Good records in every stage, which is suitable to academic reporting and assessment.
- Works with the requirements that may appear clear at the start of the project.

Why Waterfall was not chosen

- All your project needs (AI behavior, tool chaining, UI) will change probably as you play around.
- Revising the previous stages is difficult, and this is dangerous to an innovative security system.
- You require feedback, testing and improvement regularly which can be best facilitated by the iterative Agile/Scrum.

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11.1.2. Agile Methodology

Agile methodology is an adaptable approach to project management through small, iterative units of work and releasing them at an appropriate frequency, followed by constant feedback and interaction amongst the team members to enable quick learning to adapt to the change. Scrum is a special agile model that puts this concept into practice by short sprints that are definitely fixed and in which the team takes high priority tasks out of a backlog and delivers a usable increment at the end of each sprint (Peek, 2024).

- The project implies research, AI integration, various tools, and thus the requirements will be changing in the course of development.
- The supervisors and other stakeholders should provide frequent feedback to improve features such as the dashboard, AI triage, and reporting.
- The team should demonstrate ongoing improvements (prototypes, partial features) rather than till the very end.

Why Agile Scrum was selected

The final decision was to use Agile Scrum since it is based on three fundamental pillars which are highly applicable in the project:

- Transparency implies that the important work and information is visible and clear to everyone within the team and stakeholders. Backlog, sprint goals, progress and Definition of Done are openly shared to eliminate concealed tasks and surprises.
- Inspection: This is the frequent examination of the product and the process in order to identify issues at an early stage. Using such events as daily stand ups, sprint reviews and backlog refinement, the team can review progress and quality thus identifying problems before they escalate.
- Adaptation is a modification of plans or methods of doing things depending on what is being inspected. Reviews and retrospective Most of the team will make adjustments to priorities, processes, or designs so that the project will be on track and constantly improving, iteration by iteration.

11.1.2.1. Justification for choosing Methodology

The best methodological choice to the project of the Intelligence Recon System is Agile Scrum due to its ability to support requirements changes, feedback, and continuous delivery of the working part.

- **The changing demands and experimentation.**
  The project will be associated with the integration of numerous security tools, tuning AI prompt and changing dashboards and reports based on the testing of the needs, and thus, the requirements cannot be entirely determined initially. The benefits of Scrum in changing scope and priorities between iterations include short sprints and backlog refinement, which do not disrupt the entire plan.
- **Early and incremental working software required.**
  The project roadmap is sprint-based (backend setup, intelligent workflow, frontend, reporting, integration) with well-defined interim and final milestones, thus there is an advantage of delivering a working increment at the end of each sprint (for e.g. backend API on the Sprint 2, intelligent workflow on the Sprint 3, UI on the Sprint 4). Scrum is specifically intended to provide a potentially shippable increment at every sprint, which is in line with the idea of showing progress during reviews and to supervisors.
- **Close cooperation, feedback and quality**
  The project demands unending feedback of supervisors and testers with regard to security workflow, AI behavior and usability, which Scrum sustains through sprint reviews and retrospectives. Agile principles also adopt the importance of continuous testing and enhancement of each sprint, which also plays an important role in a security-oriented system that reliability and correctness are paramount.

Click here to go back.

11.1.3. V-Model Methodology

V-Model is a verification/ validation SDLC model in which each development phase on the left (requirements, system design, detailed design, coding) has an equivalent testing phase on the right (unit, integration, system, acceptance testing). The rationale is to design tests in advance and make sure that each requirement is met by a test activity (GeeksForGeeks, 2025).

Why it was considered

V-Model is desirable due to the following reasons:

- The specifications are quite precise: single internet based platform, OWASP based recon, AI triage, 15 tools, reporting, etc., thus, it is possible to map each requirement to tests.
- It is a small to medium sized project and that is where V-Model is best suited and its well-written documentation and test planning can make a student project look like an academic endeavor.
- Security domain the security domain enjoys good verification: unit tests of parsers, integration tests of tool execution, and system tests of end-to-end scans are all compatible with the systematic testing focus of the V-Model.

Why it was not chosen

Nevertheless, V-Model is not perfectly appropriate to your case, and Agile/Scrum (sprint-based plan that you are applying) is more suitable:

- The requirements will tend to change: requirements will vary as you experiment, and V-model assumes that requirements will be fixed and stable, whereas the V-model is inflexible when you need to make mid-project changes.
- In pure V-Model, working software is delivered late, whereas in this project require early prototypes (backend first, then intelligent workflow, then frontend) to demonstrate progress at interim milestones and make revisions.
- V-Model is paperwork-intensive and not as flexible as can be an overhead to a one-developer student project where rapid turnaround and quick completion is more significant than phase gates.

Click here to go back.

11.2. Technical Diagram

A technical diagram in this project is a drawing depicting visually how the system will be constructed and how the components of the system will communicate. It is concerned with architecture, components, data flow and technologies but not with user steps (draw.io, 2023).

11.2.1. Flow Chart

A flow chart of this project is a step by step diagram, which displays the sequence of tasks or processes the system undertakes in a recon scan, such as common symbols as a rectangle (process), a diamond (decision), and a control flow arrow.

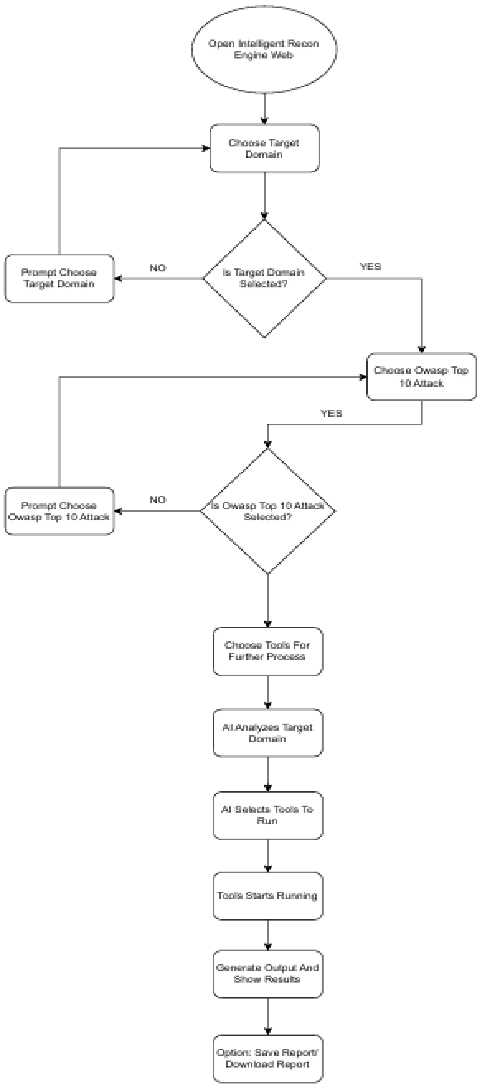

Figure 18 Flowchart Diagram

11.2.2. System architecture diagram

System architecture diagram displays HTML/CSS/JS, FastAPI, Python, and Gemini API as the frontend, API, and the 15 security tools, as well as the flow of the data between them.

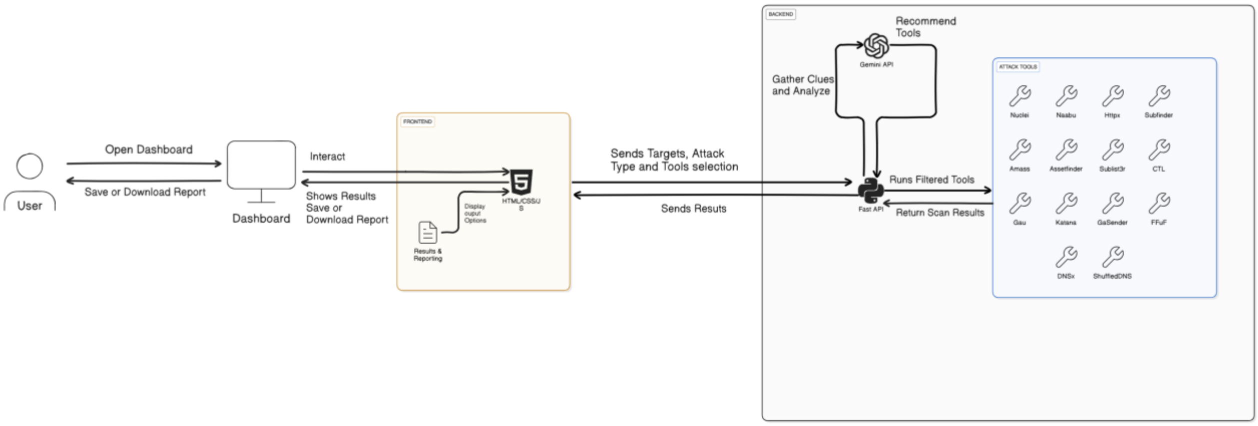

Figure 19 System Architecture Diagram